2.1.4 ▶ Ideal Gas Law

The famous Ideal Gas Law $PV = nRT$ was originally derived by experiment as an amalgamation of other emperical laws, namely Boyle's Law, Charles' Law, Gay-Lussac's Law, and Avogadro's Law. However, using the kinetic theory of gasses, it's possible to derive the law from first principles.

We start by considering a cube with side length L filled with a monatomic ideal gas. Since the gas exerts equal pressure in all directions, if we can relate the pressure of particles bouncing off of a single wall to the box's volume and temperature, we simply get the ideal gas law. For this derivation, we look at the particles bouncing off of the wall in the positive x axis (shown in blue). In a small time Δt, some particles traveling in the positive x direction collide with the blue surface, and those particles take up the volume defined by $\Delta l = \overline{v_x}\Delta t$ where $\overline{v_x}$ is the average x component of the velocities of the gas particles.

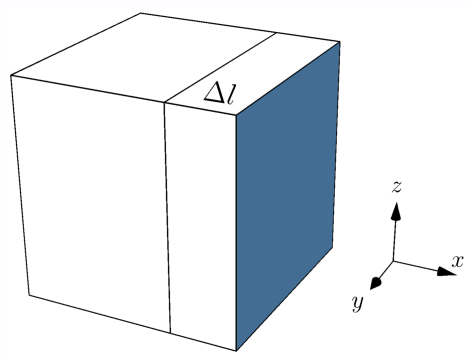

Fig. 2.2: Box with gas particles randomly moving inside. In a certain time Δt, some particles inside the section Δl hit the blue wall.

In this section of the cube, the number of particles is the number density of particles in the entire box multiplied by the volume of the section, since the particles are all evenly distributed. That number is

$$\frac{N}{V}\Delta l A = \frac{N A v_x \Delta t}{V}.$$

Of these particles that move on the x axis, half of them move towards the blue surface and half away. However, only those that move toward the surface exert any pressure, bringing our total considered particles to

$$\frac{N A v_x \Delta t}{2V}. \tag{2.16}$$

Since the gas is ideal, it collides elastically withe the walls of the container, imparting a force of $F = 2mv_x/\Delta t$ each. Thus, using eq. 2.16, the total force imparted on the blue side is

$$\frac{N A m v_x^2}{V}. \tag{2.17}$$

Since $P = F/A$, the average pressure on the blue side is

$$\langle P \rangle = \frac{N m \langle v_x^2 \rangle}{V}. \tag{2.18}$$

Since the particles are moving in all directions randomly, we can write the relation $v_{rms}^2 = \langle v^2 \rangle = \langle v_x^2 \rangle + \langle v_y^2 \rangle + \langle v_z^2 \rangle = 3\langle v_x^2 \rangle$, which implies that $\langle v_x^2 \rangle = \frac{k_B T}{m}$. Substituting everything in, we have[6]

$$P = \frac{N k_B T}{V}$$

$$PV = N k_B T \tag{2.19}$$

$$PV = nRT \tag{2.20}$$

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[6] I removed the angle brackets around P here since the P in $PV = nRT$ represents the average pressure of the gas in a certain state of the gas. More specifically, the P in $PV = nRT$ represents the "average" over all of the microstates that a given macrostate encompasses.